Local Anesthetics MCQ – Chapter 15

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 6: Regional Anesthesia (Chapter 15)

50 Questions • 35 Minutes

Instructions:
- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. Chemically, local anesthetics consist of a benzene ring separated from a tertiary amide by either an ester or amide linkage. They are accordingly classified as: **A.** Aminoesters and aminoamides **B.** Weak acids and weak bases **C.** Short and long acting **D.** Ionized and nonionized	**Ans: A**

Q2. Aminoester local anesthetics are metabolized by pseudocholinesterase and have a high incidence of allergic reactions due to: **A.** Methylparaben **B.** Para aminobenzoic acid **C.** Orthotoluidine **D.** Ecgonine	**Ans: B**

Q3. Which is the shortest acting local anesthetic? **A.** Procaine **B.** Lignocaine **C.** Chloroprocaine **D.** Dibucaine	**Ans: C**

Q4. Which is the longest acting local anesthetic? **A.** Bupivacaine **B.** Ropivacaine **C.** Tetracaine **D.** Dibucaine	**Ans: D**

Q5. Local anesthetics penetrate the axonal membrane in their undissociated (nonionized) form, then inside the cell they bind to which structure in dissociated (ionized) form? **A.** Sodium channel (alpha subunit) from the inner side **B.** Potassium channel from outside **C.** GABA receptor **D.** Calcium channel from outside	**Ans: A**

Q6. Recent studies have proven that local anesthetics not only act on sodium channels but also block which channels? **A.** Potassium and calcium channels **B.** Chloride channels **C.** Magnesium channels **D.** No other channels	**Ans: A**

Q7. According to current understanding, which nerve fibers are the MOST sensitive to local anesthetic block?

A. Large myelinated A alpha and A beta fibers
B. Small myelinated A gamma and A delta fibers
C. Nonmyelinated C fibers
D. B fibers

Ans: B

Q8. In peripheral nerve blocks, what is the functional sequence of blockade?

A. Autonomic > sensory > motor
B. Motor > sensory > autonomic
C. Sensory > motor > autonomic
D. All blocked simultaneously

Ans: B

Q9. Lignocaine has a faster onset than bupivacaine because:

A. It has a lower pKa (7.8) closer to physiologic pH, so more is in nonionized form to cross the membrane
B. It has a higher molecular weight
C. It is more lipid soluble
D. It has higher protein binding

Ans: A

Q10. Why is the onset of local anesthetic action delayed in ischemic or infected (abscess) tissue?

A. Acidic pH ionizes the drug, delaying onset
B. Alkaline pH ionizes the drug
C. The drug is metabolized faster
D. There is no nerve to block

Ans: A

Q11. A stimulated nerve is blocked earlier than a nonstimulated nerve. This kind of block is called:

A. Differential block
B. Use-dependent block
C. Phase II block
D. Refractory block

Ans: B

Q12. The most commonly used vasoconstrictor added to local anesthetics is adrenaline, in a concentration of:

A. 1 in 2,00,000
B. 1 in 1,000
C. 1 in 10,000
D. 1 in 50,000

Ans: A

Q13. The maximum systemic absorption of local anesthetic is seen after which block?

A. Spinal block
B. Intercostal nerve block
C. Axillary block
D. Infiltration

Ans: B

Q14. The potency of a local anesthetic directly correlates with its: **A.** Lipid solubility **B.** pKa **C.** Protein binding **D.** Molecular weight	**Ans: A**
Q15. The CNS:CVS dose ratio for lignocaine is 1:7 and for bupivacaine is 1:3. What does this indicate? **A.** With bupivacaine, CNS toxicity is involved at much lower doses compared to CVS toxicity (less safety margin) **B.** Bupivacaine is safer than lignocaine **C.** Lignocaine causes more cardiac toxicity **D.** Both have equal toxicity	**Ans: A**
Q16. What is often the first presentation of local anesthetic toxicity? **A.** Cardiac arrest **B.** Convulsions **C.** Hypotension **D.** Bradycardia	**Ans: B**
Q17. At clinically used doses, all local anesthetics are vasodilators EXCEPT which agents? **A.** Cocaine, levobupivacaine and ropivacaine **B.** Lignocaine and bupivacaine **C.** Procaine and tetracaine **D.** Prilocaine and mepivacaine	**Ans: A**
Q18. Allergic reactions to local anesthetics are common with esters but rare with amides. The reaction with amides is because of which preservative? **A.** Methylparaben **B.** Sodium bisulfite **C.** Para aminobenzoic acid **D.** Benzethonium chloride	**Ans: A**
Q19. Methemoglobinemia is usually seen with which local anesthetics? **A.** Prilocaine and benzocaine **B.** Lignocaine and bupivacaine **C.** Cocaine and procaine **D.** Tetracaine and dibucaine	**Ans: A**
Q20. Local anesthetics with adrenaline can cause necrosis and gangrene if used for ring block of which structures (because these have end arteries)? **A.** Fingers, toes, penis or pinna **B.** Forearm and leg **C.** Abdomen and back **D.** Scalp and forehead	**Ans: A**

Q21. Which was the first local anesthetic used, and what is its source? **A.** Procaine, from coca leaves **B.** Cocaine, extracted from the leaves of Erythroxylum coca **C.** Lignocaine, synthesized chemically **D.** Tetracaine, from cinchona bark	**Ans: B**

Q22. Which local anesthetic is the agent of choice in patients with a history of malignant hyperthermia? **A.** Lignocaine **B.** Bupivacaine **C.** Procaine **D.** Cocaine	**Ans: C**

Q23. Which is the most commonly used local anesthetic? **A.** Bupivacaine **B.** Lignocaine (xylocaine, lidocaine) **C.** Procaine **D.** Ropivacaine	**Ans: B**

Q24. What is the maximum safe dose of lignocaine without and with adrenaline? **A.** 4.5 mg/kg (max 300 mg) without adrenaline; 7 mg/kg (max 500 mg) with adrenaline **B.** 2.5 mg/kg without; 3 mg/kg with adrenaline **C.** 7 mg/kg without; 4.5 mg/kg with adrenaline **D.** 3 mg/kg without; 5 mg/kg with adrenaline	**Ans: A**

Q25. Lignocaine is used intravenously for the treatment of ventricular tachycardia. The preparation used is xylocard 2%, in a dose of: **A.** 2 mg/kg **B.** 5 mg/kg **C.** 0.5 mg/kg **D.** 10 mg/kg	**Ans: A**

Q26. Which local anesthetic is the safest, owing to its extrahepatic metabolism (in liver, kidneys, lungs by amidase)? **A.** Lignocaine **B.** Prilocaine **C.** Bupivacaine **D.** Mepivacaine	**Ans: B**

Q27. Bupivacaine is the local anesthetic of choice for postoperative pain relief and painless labor because of its: **A.** Wide differential sensory and motor blockade (motor block only at high concentrations) **B.** Short duration of action **C.** Rapid onset **D.** Low protein binding	**Ans: A**

Q28. What is the maximum safe dose of bupivacaine without and with adrenaline?

- **A.** 2.5 mg/kg (max 175 mg) without; 3 mg/kg (max 225 mg) with adrenaline
- **B.** 4.5 mg/kg without; 7 mg/kg with adrenaline
- **C.** 3 mg/kg without; 4 mg/kg with adrenaline
- **D.** 1 mg/kg without; 2 mg/kg with adrenaline

Ans: A

Q29. Bupivacaine is absolutely contraindicated for which block due to its high cardiotoxicity?

- **A.** Spinal block
- **B.** Bier's block (IVRA)
- **C.** Epidural block
- **D.** Nerve block

Ans: B

Q30. Which drug is now considered the drug of choice for the treatment of bupivacaine (and other local anesthetic) induced ventricular tachycardia?

- **A.** Bretylium
- **B.** Amiodarone
- **C.** Lignocaine
- **D.** Procainamide

Ans: B

Q31. Levobupivacaine is the S isomer of bupivacaine. Why is it less cardiotoxic than bupivacaine?

- **A.** Because it does not contain the R isomer
- **B.** Because it is more lipid soluble
- **C.** Because it is metabolized faster
- **D.** Because it has lower protein binding

Ans: A

Q32. Ropivacaine cardiotoxicity compared to bupivacaine and levobupivacaine is:

- **A.** Less than both bupivacaine and levobupivacaine (but still higher than lignocaine)
- **B.** More than both
- **C.** Equal to bupivacaine
- **D.** More than lignocaine but less than bupivacaine only

Ans: A

Q33. Why is ropivacaine always preferred over bupivacaine for postoperative analgesia and painless labor?

- **A.** It is less potent and produces less motor block, while sensory analgesia is comparable
- **B.** It has a faster onset
- **C.** It is cheaper
- **D.** It has a longer duration

Ans: A

Q34. EMLA cream is a eutectic mixture of which two local anesthetics?

- **A.** 5% prilocaine and 5% lignocaine in equal amount (1:1 ratio)
- **B.** Tetracaine and adrenaline
- **C.** Bupivacaine and ropivacaine
- **D.** Cocaine and tetracaine

Ans: A

Q35. Which local anesthetics CANNOT be used for surface (topical) anesthesia as they do not get absorbed from skin or mucous membranes? **A.** Mepivacaine, bupivacaine, levobupivacaine and ropivacaine **B.** Lignocaine and prilocaine **C.** Cocaine and tetracaine **D.** Benzocaine and amethocaine	**Ans: A**
Q36. Which local anesthetic concentration is used for intravenous regional analgesia (Bier's block)? **A.** 5% lignocaine **B.** 0.5% lignocaine **C.** 2% lignocaine **D.** 0.125% bupivacaine	**Ans: B**
Q37. Cardiotoxicity of bupivacaine increases in which conditions? **A.** Pregnancy, hypoxia and acidosis **B.** Hyperventilation and alkalosis **C.** Hypothermia only **D.** Hyperkalemia only	**Ans: A**
Q38. Local anesthetic induced neurotoxicity classically manifests as cauda equina syndrome, seen with which agents through small bore catheters? **A.** Lignocaine and tetracaine **B.** Bupivacaine and ropivacaine **C.** Procaine and cocaine **D.** Mepivacaine and prilocaine	**Ans: A**
Q39. For spinal and epidural anesthesia, which type of local anesthetic preparation should be used? **A.** Preservative free preparations **B.** Preparations with methylparaben **C.** Preparations with adrenaline **D.** Acidic preparations (pH 4)	**Ans: A**
Q40. Tumescent anesthesia is a technique used by plastic surgeons during liposuction. What is a concern with this technique? **A.** Peak concentration may occur in postoperative period producing toxicity **B.** It causes immediate methemoglobinemia **C.** It is contraindicated in all patients **D.** It requires general anesthesia	**Ans: A**
Q41. Which agent is used for refrigeration analgesia, where nerve conduction is blocked by cooling the nerves to –5 to –20°C? **A.** Cryoprobes delivering CO_2 or nitrous oxide **B.** Liquid nitrogen **C.** Ice packs **D.** Ethyl chloride spray	**Ans: A**

Q42. When two local anesthetics are combined (e.g. lignocaine for rapid onset and bupivacaine for long duration), the toxicity is:

A. Additive (not independent), therefore maximum safe doses should be reduced accordingly
B. Independent of each other
C. Reduced
D. Neutralized

Ans: A

Q43. Differential block of local anesthetics depends on concentration. A drug at lower concentration will:

A. Produce only sensory block, while at higher concentration produce motor block
B. Produce only motor block
C. Produce no block
D. Block autonomic fibers only

Ans: A

Q44. Which agent has the property of being the longest acting, most potent, and most toxic local anesthetic?

A. Bupivacaine
B. Tetracaine
C. Dibucaine (cinchocaine)
D. Ropivacaine

Ans: C

Q45. Procaine and chloroprocaine are metabolized by pseudocholinesterase. Cocaine differs in that it is metabolized in:

A. Liver
B. Plasma
C. Lungs
D. Kidneys

Ans: A

Q46. Concentration minimum (Cm) — the minimum concentration of local anesthetic to block nerve impulse conduction — is increased by:

A. Acidic pH (causing ionization)
B. Alkaline pH
C. Nerve stimulation
D. Adjuvant opioids

Ans: A

Q47. In central neuraxial blocks, the sensory sequence of blockade differs from peripheral nerve blocks regarding temperature and pain. In central neuraxial blocks:

A. Temperature is blocked earlier than pain
B. Pain is blocked earlier than temperature
C. Touch is blocked first
D. Proprioception is blocked first

Ans: A

Q48. What is the most common cause of failure of local anesthesia?

A. Inappropriate technique, inadequate volume or concentration
B. Drug allergy
C. Genetic variation
D. Tachyphylaxis

Ans: A

Q49. Adrenaline added to commercial local anesthetic preparations makes them even more acidic (pH around 4). Why? **A.** Because adrenaline can become unstable at alkaline pH **B.** To increase potency **C.** To reduce allergic reactions **D.** To improve shelf life	**Ans: A**

Q50. Which opioid is frequently used to supplement the effect of local anesthetics for nerve blocks because of its local anesthetic properties? **A.** Morphine **B.** Buprenorphine **C.** Fentanyl **D.** Remifentanil	**Ans: B**

Answer Key & Explanations

Q1. Answer: A — Aminoesters and aminoamides

Page 151: Local anesthetics consist of a benzene ring separated from a tertiary amide either by ester or amide linkage. Based on this intermediate chain they are classified as aminoesters and aminoamides.

Topic: Classification

Q2. Answer: B — Para aminobenzoic acid

Page 151, Table: Aminoesters are metabolized by pseudocholinesterase (except cocaine which is metabolized in liver) and have a high incidence of allergic reactions due to para aminobenzoic acid. Amides are metabolized primarily in liver and have a low incidence of allergic reactions.

Topic: Classification

Q3. Answer: C — Chloroprocaine

Page 151: Based on duration of action, short duration (15–30 minutes) agents are chloroprocaine (shortest acting local anesthetic) and procaine. Dibucaine is the longest acting local anesthetic.

Topic: Classification

Q4. Answer: D — Dibucaine

Page 151: Long duration (2–3 hours) agents include bupivacaine, levobupivacaine, ropivacaine, tetracaine, etidocaine and dibucaine. Dibucaine is the longest acting local anesthetic.

Topic: Classification

Q5. Answer: A — Sodium channel (alpha subunit) from the inner side

Page 152: Local anesthetics in undissociated (nonionized) form penetrate the axonal membrane. Once inside, the dissociated (ionized, protonated, cationic) form binds to the sodium channel (alpha subunit) from the inner side, blocking the channel and preventing depolarization and action potential.

Topic: Mechanism of Action

Q6. Answer: A — Potassium and calcium channels

Page 152: Recent studies have proven that local anesthetics not only act on sodium channels, but also block potassium and calcium channels in a manner that hypokalemia and hypercalcemia can antagonize the effect of local anesthetics. They are also found to block N-methyl-D-aspartate (NMDA) receptors.

Topic: Mechanism of Action

Q7. Answer: B — Small myelinated A gamma and A delta fibers

Page 152: Small myelinated fibers (A gamma and A delta) are the most sensitive to local anesthetic block, followed by large myelinated (A alpha and A beta), then B (mixed), with nonmyelinated C fibers being most resistant. The sequence of blockade is A > B > C. The traditional notion that thin fibers are most sensitive does not hold true in present day practice.

Topic: Sensitivity of Nerve Fiber

Q8. Answer: B — Motor > sensory > autonomic

Page 153: In peripheral nerve blocks, motor function is blocked earliest followed by sensory and then autonomic (motor > sensory > autonomic). However, in central neuraxial blocks the sequence is autonomic > sensory > motor. Sequence of recovery is in reverse order of blockade.

Topic: Sensitivity of Nerve Fiber

Q9. Answer: A — It has a lower pKa (7.8) closer to physiologic pH, so more is in nonionized form to cross the membrane

Page 153: pKa is the pH at which a local anesthetic is 50% ionized and 50% nonionized. Agents with pKa closer to physiologic pH have more nonionized form which diffuses through the axonal membrane, enhancing onset. That is why lignocaine with lower pKa of 7.8 has fast onset compared to bupivacaine with higher pKa of 8.1.

Topic: Onset of Action

Q10. Answer: A — Acidic pH ionizes the drug, delaying onset

Page 153: Adding sodabicarbonate makes more drug available in nonionized form to cross the axonal membrane (faster onset). On the other hand, in ischemic tissue (like abscess) acidic pH ionizes the drug, delaying the onset of action.

Topic: Onset of Action

Q11. Answer: B — Use-dependent block

Page 153: Since activated channels are blocked more easily, a stimulated nerve will be blocked earlier than a nonstimulated nerve. This kind of block by local anesthetics is called use-dependent block.

Topic: Onset of Action

Q12. Answer: A — 1 in 2,00,000

Page 153: Vasoconstrictors decrease the systemic absorption increasing the duration of action. The most commonly used vasoconstrictor is adrenaline in a concentration of 1 in 2,00,000. Adrenaline when added to lignocaine increases the duration of both sensory and motor blockade, while with bupivacaine only sensory block is prolonged.

Topic: Duration of Action

Q13. Answer: B — Intercostal nerve block

Page 153: Systemic absorption depends on the site of injection, proportional to the vascularity of the site. Maximum systemic absorption has been seen after intercostal nerve block. Addition of vasoconstrictors decreases the systemic absorption.

Topic: Systemic Absorption

Q14. Answer: A — Lipid solubility

Page 153: Potency directly correlates with lipid solubility. Potency (lipid solubility) is also directly proportional to duration; more the potency, longer is the duration (with the exception of chloroprocaine which despite having intermediate potency is shortest acting).

Topic: Potency

Q15. Answer: A — With bupivacaine, CNS toxicity is involved at much lower doses compared to CVS toxicity (less safety margin)

Page 154: The CNS:CVS dose ratio for lignocaine is 1:7 and for bupivacaine is 1:3, meaning that with bupivacaine, CNS toxicity is involved at much lower doses as compared to CVS — indicating a narrower safety margin. CNS is the first system involved in local anesthetic toxicity.

Topic: Systemic Toxicity

Q16. Answer: B — Convulsions

Page 154: As the CNS is the first system involved in local anesthetic toxicity, typical sequence is excitation followed by depression of cerebral tissue. Signs include circumoral numbness, dizziness, tongue paresthesia, visual and auditory disturbances, muscle twitching, tremors, convulsions followed by coma and death. Convulsions as first presentation are quite common.

Topic: Systemic Toxicity

Q17. Answer: A — Cocaine, levobupivacaine and ropivacaine

Page 154: At lower doses local anesthetics may act as vasoconstrictors by directly inhibiting nitric oxide, but at clinically used doses all local anesthetics are vasodilators except cocaine, levobupivacaine and ropivacaine which are vasoconstrictors at all doses.

Topic: Cardiovascular System

Q18. Answer: A — Methylparaben

Page 154–155: Allergic reactions are common with esters but rare with amides. The reaction with amides is because of the preservative (methyl paraben) it contains. Cross sensitivity does not exist between classes (i.e. esters and amides) but exists between agents of the same class.

Topic: Immunologic

Q19. Answer: A — Prilocaine and benzocaine

Page 155: Methemoglobinemia is usually seen with prilocaine; however, benzocaine can also cause methemoglobinemia. Treatment is IV methylene blue.

Topic: Systemic Effects

Q20. Answer: A — Fingers, toes, penis or pinna

Page 155: Local anesthetics with adrenaline can cause necrosis and gangrene if used for ring block of fingers, toes, penis or pinna because these structures have end arteries.

Topic: Local Toxicity

Q21. Answer: B — Cocaine, extracted from the leaves of Erythroxylum coca

Page 155: Cocaine was the first local anesthetic used by Carl Koller for anesthetizing the cornea. It is extracted from the leaves of Erythroxylum coca. It is a very potent vasoconstrictor and is the only ester not metabolized by pseudocholinesterase (it is metabolized in liver).

Topic: Individual Agents

Q22. Answer: C — Procaine

Page 155: Procaine, like other esters, is metabolized by pseudocholinesterase. It is the agent of choice in patients with history of malignant hyperthermia. In fact, historically it had been used for the treatment of malignant hyperthermia.

Topic: Individual Agents

Q23. Answer: B — Lignocaine (xylocaine, lidocaine)

Page 155: Lignocaine (xylocaine, lidocaine) is the most commonly used local anesthetic. It was first synthesized by Lofgren and first used by Gordh. Its solution is very stable, not even decomposed by boiling.

Topic: Individual Agents

Q24. Answer: A — 4.5 mg/kg (max 300 mg) without adrenaline; 7 mg/kg (max 500 mg) with adrenaline

Page 156: The maximum safe dose of lignocaine without adrenaline is 4.5 mg/kg (maximum 300 mg) and with adrenaline is 7 mg/kg (maximum 500 mg). Duration of effect without adrenaline is 45–60 minutes and with adrenaline is 2–3 hours.

Topic: Lignocaine

Q25. Answer: A — 2 mg/kg

Page 156: For treating ventricular tachycardia, preservative free lignocaine (available as xylocard 2%) is used intravenously in a dose of 2 mg/kg. Intravenous xylocard is also used for blunting the cardiovascular response to laryngoscopy and intubation, and lignocaine infusion is used for neuropathic pain.

Topic: Lignocaine

Q26. Answer: B — Prilocaine

Page 156: Prilocaine, other than liver, also has extrahepatic metabolism in kidneys, lungs and by amidase, making it the safest local anesthetic. Methemoglobinemia occurs with prilocaine at high doses, because of accumulation of its metabolite orthotoluidine which converts hemoglobin to methHb.

Topic: Prilocaine

Q27. Answer: A — Wide differential sensory and motor blockade (motor block only at high concentrations)

Page 156: The very unique property of bupivacaine is wide differential sensory and motor blockade, which means motor block will occur only at high concentrations, making it a local anesthetic of choice for postoperative pain relief and painless labor. It is 4 times more potent than lignocaine.

Topic: Bupivacaine

Q28. Answer: A — 2.5 mg/kg (max 175 mg) without; 3 mg/kg (max 225 mg) with adrenaline

Page 156: The maximum safe dose of bupivacaine without adrenaline is 2.5 mg/kg (maximum 175 mg) and with adrenaline is 3 mg/kg (maximum 225 mg). Addition of adrenaline only prolongs the duration of sensory block.

Topic: Bupivacaine

Q29. Answer: B — Bier's block (IVRA)

Page 157: Bupivacaine has high tissue binding (slow reversal of sodium channels) and high degree of protein binding which makes resuscitation after cardiac arrest prolonged and very difficult. Therefore, it is absolutely contraindicated for Bier's block.

Topic: Bupivacaine

Q30. Answer: B — Amiodarone

Page 157: Contrary to the previous recommendation of bretylium as the drug of choice, amiodarone nowadays is considered as the drug of choice for treatment of bupivacaine (and other local anesthetic) induced ventricular tachycardia. Management of cardiac arrest includes CPR along with rapid bolus of Intralipid 20%, 1.5 mL/kg.

Topic: Bupivacaine

Q31. Answer: A — Because it does not contain the R isomer

Page 157: Levobupivacaine is the S isomer of bupivacaine. As it does not contain the R isomer (which is responsible for cardiotoxicity), the cardiotoxicity of levobupivacaine is less than bupivacaine. Maximum safe dose of levobupivacaine is 2.5 mg/kg (maximum 175 mg).

Topic: Levobupivacaine

Q32. Answer: A — Less than both bupivacaine and levobupivacaine (but still higher than lignocaine)

Page 157: Cardiotoxicity and CNS toxicity of ropivacaine is less as compared to bupivacaine and even lesser than levobupivacaine (but still higher than lignocaine). Maximum safe dose is 3 mg/kg (maximum 225 mg). Since its cardiotoxicity is less, it can be given in higher doses.

Topic: Ropivacaine

Q33. Answer: A — It is less potent and produces less motor block, while sensory analgesia is comparable

Page 157: Ropivacaine is less potent than bupivacaine, therefore its duration and intensity of motor block will also be proportionately less, while sensory analgesia is comparable. Therefore, ropivacaine is always preferred over bupivacaine for postoperative analgesia and painless labor (also being less cardiotoxic).

Topic: Ropivacaine

Q34. Answer: A — 5% prilocaine and 5% lignocaine in equal amount (1:1 ratio)

Page 157: EMLA stands for eutectic (easily melted) mixture of local anesthetics. It is a combination of 5% prilocaine and 5% lignocaine in equal amount (1:1 ratio). It is used for intravenous cannulation in children or sensitive individuals, small skin procedures and removal of mole. The major limitation is its slow onset (30–60 min).

Topic: Methods of Local Anesthesia

Q35. Answer: A — Mepivacaine, bupivacaine, levobupivacaine and ropivacaine

Page 157: Mepivacaine, bupivacaine, levobupivacaine and ropivacaine do not have any absorption from skin or mucous membranes therefore cannot be used for topical anesthesia. Also, the absorption of procaine and chloroprocaine is so poor that they should not be used for surface anesthesia.

Topic: Methods of Local Anesthesia

Q36. Answer: B — 0.5% lignocaine

Page 156, 158: 0.5% lignocaine is used for Bier's block (intravenous regional analgesia). Spinal uses 5% heavy lignocaine, nerve blocks 1–2%, surface (topical) 4% and 10%, and infiltration 1–2%.

Topic: Methods of Local Anesthesia

Q37. Answer: A — Pregnancy, hypoxia and acidosis

Page 156: Cardiotoxicity of bupivacaine increases in pregnancy, hypoxia and acidosis. It is far more cardiotoxic than lignocaine, with the R component contributing more than the S component in causing cardiotoxicity.

Topic: Bupivacaine

Q38. Answer: A — Lignocaine and tetracaine

Page 155: The classical example of local anesthetic induced local neurotoxicity is cauda equina syndrome seen with lignocaine and tetracaine if used through small bore spinal catheters. Transient neurologic symptoms (dysesthesia, radicular pain in lower extremities) are seen with lignocaine and mepivacaine. Maximum local neurotoxicity has been reported with chloroprocaine.

Topic: Local Toxicity

Q39. Answer: A — Preservative free preparations

Page 155: Antimicrobial preservative (methylparaben) is added to multidose vials. For spinal and epidural anesthesia, preservative free preparations should be used.

Topic: Commercial Preparations

Q40. Answer: A — Peak concentration may occur in postoperative period producing toxicity

Page 158: Tumescent anesthesia is used by plastic surgeons during liposuction, where dilute combination of lignocaine with adrenaline is injected subcutaneously in large quantities. Peak concentration may even occur in postoperative period producing toxicity.

Topic: Methods of Local Anesthesia

Q41. Answer: A — Cryoprobes delivering CO_2 or nitrous oxide

Page 158: In refrigeration analgesia, nerve conduction is blocked by cooling the nerves using cryoprobes which deliver CO_2 or nitrous oxide at –5 to –20°C.

Topic: Methods of Local Anesthesia

Q42. Answer: A — Additive (not independent), therefore maximum safe doses should be reduced accordingly

Page 155: The use of two local anesthetics is quite common in clinical practice (usually lignocaine for rapid onset and bupivacaine/ropivacaine for long duration). The toxicity of two local anesthetics should be considered additive, not independent, and therefore maximum safe doses should be reduced accordingly.

Topic: Toxicity

Q43. Answer: A — Produce only sensory block, while at higher concentration produce motor block

Page 154: Differential block depends on concentration. A drug at lower concentration will produce only sensory block while at higher concentration can produce motor block.

Topic: Differential Block

Q44. Answer: C — Dibucaine (cinchocaine)

Page 157, Table 15.2: Dibucaine (cinchocaine) is longest acting, most potent and most toxic local anesthetic.

Topic: Individual Agents

Q45. Answer: A — Liver

Page 154–155: Esters are metabolized by plasma pseudocholinesterase except cocaine which is metabolized in liver. Cocaine is the only ester not metabolized by pseudocholinesterase. Its metabolite ecgonine is also a CNS stimulant.

Topic: Metabolism

Q46. Answer: A — Acidic pH (causing ionization)

Page 154: Concentration minimum (Cm) is the minimum concentration of local anesthetic that will block nerve impulse conduction. Acidic pH increases the Cm by causing ionization. Stimulated nerve will need less Cm. Hypokalemia and hypercalcemia antagonize the block.

Topic: Concentration Minimum

Q47. Answer: A — Temperature is blocked earlier than pain

Page 153: The sequence of sensory blockade is pain > temperature (cold before hot) > touch > deep pressure > proprioception, EXCEPT for central neuraxial blocks where temperature is blocked earlier than pain.

Topic: Sensitivity of Nerve Fiber

Q48. Answer: A — Inappropriate technique, inadequate volume or concentration

Page 158: The usual cause of failure of local anesthesia is inappropriate technique, inadequate volume or concentration of local anesthetic. However, certain factors like infection (acidosis), genetic variation, or drug tolerance and tachyphylaxis can also contribute to failure.

Topic: Causes of Failure

Q49. Answer: A — Because adrenaline can become unstable at alkaline pH

Page 155: Local anesthetics are made acidic (pH around 6) to enhance chemical stability. Local anesthetics containing adrenaline are made even more acidic (pH around 4) because adrenaline can become unstable at alkaline pH.

Topic: Commercial Preparations

Q50. Answer: B — Buprenorphine

Page 151: Among other drugs with local anesthetic properties, buprenorphine — because of its local anesthetic properties — is frequently used to supplement the effect of local anesthetics for nerve blocks. Pethidine and tramadol also have local anesthetic properties.

Topic: Other Drugs with LA Properties